NLP Location Identification Accuracy
Using LLMs to identify locations in unclean chat messages.

Natalie Crowell
Rochester Institute of Technology
nec9907@rit.edu

Olivia Croteau
Rochester Institute of Technology
osc3732@rit.edu

Abstract

The growing volume of unstructured communication data from displaced populations presents both challenges and opportunities for humanitarian organizations seeking to understand community needs and allocate resources effectively. This paper presents a comparative study of two machine learning approaches for automated location extraction and geocoding from chat message data: a BERT-based Named Entity Recognition model and GazPNE, a gazetteer-leveraged deep learning method. We evaluate both models on their ability to identify location entities in unstructured text and resolve them to geographic coordinates using a comprehensive gazetteer covering Ukraine, Poland, and Russia. Our BERT model achieved an F1-score of 0.93 for location extraction with strong precision (0.93), while GazPNE demonstrated exceptional recall (0.97) and significantly faster processing time (7.03 seconds vs. 114.82 seconds). Geocoding evaluation on real Telegram message data from Ukrainian refugee communities revealed practical challenges including gazetteer coverage limitations and toponym ambiguity. The BERT model successfully geocoded 26 of 92 extracted locations, while GazPNE resolved 20 of 2,391 extractions, highlighting the trade-off between precision and recall in real-world applications. Our findings demonstrate that both approaches offer viable solutions for humanitarian data analysis, with model selection depending on specific operational requirements for accuracy versus processing speed. This research contributes practical tools and insights for spatial analysis of crisis communication data.

1 Introduction

The growing use of digital communication platforms has created large volumes of unstructured text data containing valuable geographic information that could support humanitarian work. This research addresses the challenge of automatically extracting location references from unstructured text and converting them to geographic coordinates. The motivation for this study stems from our previous research analyzing Telegram messages from Ukrainian refugees in Poland (to be published), which highlighted the need for systematic methods to identify locations of importance to displaced communities. By developing effective geoparsing methods this research aims to help humanitarian data analysts and decision makers understand where refugee communities are gathered and what concerns are being raised in specific locations. The ability to map message data enables humanitarian organizations to make informed decisions about resource allocation and service delivery based on spatial patterns of community needs.

This study addresses two primary research questions: (1) To what extent can machine learning models accurately identify location entities in unstructured chat message data, as measured by precision, recall, and F1-score on test data? (2) How accurately can identified toponyms be geocoded to their correct latitude and longitude coordinates compared to ground truth coordinates? By addressing these questions, this research advances automated geoparsing methods for multilingual crisis communication while providing practical tools for humanitarian operations.

2 Related Work

Three prior studies significantly informed our methodological approach. First, "Location Reference Recognition from Texts: A Survey and Comparison" published by the Association for Computing Machinery (Hu et al., 2023) provides a comprehensive evaluation of location recognition models using standardized metrics. This survey guided our selection of evaluation metrics and identified promising models for implementation. Notably, the authors highlight the GazPNE model, which "fuses rules, gazetteers, and deep learning methods without requiring any manually annotated data" and can extract place names at multiple granularity levels, including abbreviated forms (Hu et al., 2023). The GazPNE approach is particularly relevant for our application due to its ability to handle informal text without manual annotation requirements.

Two additional studies informed our second methodological approach. "BERT4Loc: BERT for Location—POI Recommender System" (Bashir et al., 2023) demonstrates a BERT-based framework for location identification that achieves strong performance on social media text, which shares characteristics with our chat message data such as informal language and abbreviated references. "Adaptive Geoparsing Method for Toponym Recognition and Resolution in Unstructured Text" (Aldana-Bobadilla et al., 2020) addresses the challenge of toponym disambiguation—resolving ambiguous place names to their correct geographic coordinates after initial extraction. Together, these studies provide foundational approaches for both the location extraction and geocoding components of our geoparsing pipeline.

3 Methods

Our proposed method compares two location extraction models. The first is a BERT-based Named Entity Recognition (NER) model that requires IOB-tagged training data, where tags indicate the beginning (B), inside (I), or outside (O) of location entities. We used the "bert-base-cased" pretrained model as our foundation, leveraging its case sensitivity to assist with identifying locational proper nouns. The model architecture includes BERT's transformer layers for contextual word embeddings and a classification layer that predicts IOB tags for unseen data. The training dataset contains 14,041 sentences from a location-focused CoNLL dataset (John, 2022).

Our second model is GazPNE (Uhuohuy, n.d.), a more complex approach that combines gazetteers, rules, and deep learning without requiring manually annotated data. This model uses binary classification with positive examples (actual place names) and negative examples (non-place terms). Positive examples were extracted from an OpenStreetMap gazetteer (MapTiler AG, 2025) covering Ukraine, Russia, and Poland, along with generic locational terms such as "Street," "River," and "Road." Negative examples were generated through techniques including combining common words and extracting individual words from multi-word place names. The model applies weighted cross-entropy loss to balance the learning from both example types.

GazPNE creates composite embeddings from three sources: GloVe word vectors, OSM gazetteer features specific to our regions of interest, and statistical features derived from positive examples (e.g., word frequency in place names and typical place name length). The model architecture processes these embeddings through a CNN layer to identify sequential patterns, an LSTM layer to analyze contextual relationships, and a classification layer to determine whether each word represents a location. The CNN component is particularly effective at capturing n-gram patterns in multi-word locations.

After location extraction, both models apply geocoding methods to assign coordinates. The BERT model uses dynamic context disambiguation, which applies rule-based logic and considers previously resolved locations to determine the most likely coordinates for ambiguous place names. The GazPNE model resolves ambiguity by prioritizing locations based on population size and feature importance. Both geocoding approaches rely on the same GeoNames gazetteer (GeoNames, n.d.) for matching extracted locations to known places with coordinate data.

4 Experiments

Our experimental setup leverages multiple datasets for training and evaluation. The BERT-based NER model was fine-tuned using a location-focused CoNLL dataset (John, 2022) containing 14,041 training sentences, 3,453 test sentences, and 3,250 validation sentences with IOB tags indicating location entities.

The GazPNE model was trained using 1,644,361 positive examples and 179,916,430 negative examples. We adapted the preprocessing script from the original GazPNE repository (Uhuohuy, n.d.), which was designed to extract examples from OSM data for the United States and India, to generate training examples specific to Ukraine, Russia, and Poland using OpenStreetMap data (MapTiler AG, 2025).

We evaluate both models at the entity level, where metrics are calculated based on complete location entities rather than individual tokens. Entity-level evaluation is more appropriate for location extraction since place names frequently span multiple words. We calculate three standard metrics: precision (the proportion of predicted locations that are correct), recall (the proportion of actual locations that are identified), and F1-score (the harmonic mean of precision and recall). Our target performance range is an F1-score between 0.90 and 0.99.

To ensure fair comparison, we created a unified test set from the CoNLL dataset (John, 2022). The BERT model was evaluated using the original IOB-tagged test sentences, while the GazPNE model required binary classification examples. We applied the preprocessing methodology to generate 505 positive and 505 negative examples from the 3,453 test sentences, maintaining consistency in the underlying text data across both evaluation approaches.

For geocoding evaluation, both models use the same GeoNames gazetteer (GeoNames, n.d.) to match extracted location references to coordinates. This gazetteer contains 508,720 locations across Ukraine, Poland, and Russia, including cities, towns, geographic features, and their alternate names and spellings.

To evaluate real-world performance, we applied our geoparsing pipeline to actual Telegram message data collected from Ukrainian refugee channels. Since this data lacks ground truth annotations, we conducted manual evaluation on a smaller subset. This qualitative assessment allows us to identify inconsistencies in model outputs, recognize patterns in entity extraction, detect outliers, and understand the challenges our models face when processing authentic unstructured communication data.

5 Results

Table 1 presents the entity-level performance of both models on the test dataset. The BERT model achieved higher precision (0.93) and F1-score (0.93), indicating greater overall accuracy in location identification. However, it required substantially longer processing time at 114.82 seconds. The GazPNE model demonstrated exceptional recall (0.97), successfully identifying 97% of all location entities, while processing the same data in only 7.03 seconds. With an F1-score of 0.90, GazPNE may be preferred in applications requiring rapid processing, though BERT offers superior precision when accuracy is the primary concern.

Table 2 shows the geocoding performance on real Telegram message data. Due to the lack of ground truth annotations for this dataset, we conducted manual evaluation on a smaller subset. The results reveal distinct challenges for each model. GazPNE extracted an extremely large number of location candidates (2,391), indicating a high false positive rate. Manual analysis showed that GazPNE frequently misidentified proper nouns, particularly personal names, as locations. The model's reliance on gazetteer matching to filter predictions was evident, with only 75 of the 2,391 extracted entities successfully resolved to coordinates.

The BERT model demonstrated more conservative extraction, identifying 92 location entities. However, geocoding challenges emerged during the resolution phase. A recurring error involved assigning coordinates to the common noun "home," which appeared in our gazetteer as an actual place name. Unresolved entities often included specific addresses such as "Zamenhofa Street" or "Ogrodowa Street," suggesting that gazetteer coverage limitations hindered geocoding accuracy. Additionally, the model frequently failed to recognize complete addresses and occasionally extracted partial location names (e.g., tagging "Royal Lazienki" when the full reference was "Royal Lazienki park"). These error patterns indicate that both models would benefit from expanded training data including more fine-grained location types and a more comprehensive gazetteer covering street-level features.

6 Ethical Considerations

This research involves the analysis of communication data collected from public Telegram channels dedicated to Ukrainian refugee communities through automated scraping methods without explicit informed consent from individual users. We acknowledge the ethical complexities of analyzing communication from vulnerable displaced populations during a humanitarian crisis.

To protect user privacy and minimize potential harm, we implemented several safeguards. No personal user metadata such as usernames, user IDs, phone numbers, or profile information were collected or stored. Our analysis focused exclusively on message text content to extract location references and assess geoparsing performance. All data were drawn from public channels accessible to any Telegram user, not private groups or direct mes-

Model	Precision	Recall	F1-Score	Testing Runtime
BERT	0.9327	0.9305	0.9316	114.82 seconds
GazPNE	0.8422	0.9723	0.9026	7.03 seconds

Table 1: Entity-level metrics of each model on the same testing data after training.

Model	Ttl. Extracted	Ttl. Resolved	Ttl. Correct
BERT	92	33	26
GazPNE	2,391	75	20

Table 2: Geoparsing results on Telegram message data. Extracted = Total locations extracted, Resolved = Total locations resolved, Correct = Correctly resolved locations.

sages.

We recognize that publicly posted messages may contain sensitive information about refugee movements and needs. However, the potential benefits of this research align with the interests of affected populations. The methods developed are designed to support, rather than surveil, displaced communities by helping humanitarian actors allocate resources more effectively.

Future applications of this work should carefully consider data governance frameworks and engage with affected communities when possible.

7 Limitations

A significant limitation of this study involves translation accuracy and its potential impact on location extraction performance. The Telegram message data were originally written in Ukrainian, Russian, and Polish, requiring translation to English before processing. This translation step was necessary because both the BERT-based NER model and GazPNE were trained on English-language datasets, and applying these models directly to multilingual text would have yielded poor results.

We employed the Gemma3-translator model from Ollama, a 4-billion parameter translation model that allows for specific instructions via modelfiles. This model was selected for its demonstrated accuracy with repeated proper nouns, which is particularly important for location extraction tasks. For example, the model consistently translated multi-word place names like "Ukrainian House" correctly across multiple message instances, maintaining the integrity of location references that are critical to our geoparsing objectives.

However, translation inevitably introduces potential sources of error. Place names may be transliterated inconsistently, culturally specific location references may lose nuance, and ambiguous terms may be translated in ways that affect subsequent entity recognition. Additionally, code-switching between languages within individual messages—common in multilingual refugee communities—presents challenges for automated translation systems. These translation-related errors compound with the inherent limitations of the location extraction and geocoding models themselves.

Future work could explore multilingual NER models trained directly on Ukrainian, Russian, and Polish text to eliminate the translation step, or develop translation quality assessment methods specifically focused on preserving location reference accuracy. Understanding the extent to which translation errors contribute to overall system performance remains an important area for further investigation.

8 Conclusion

This research developed and compared two approaches for automated location extraction and geocoding from unstructured chat message data to support humanitarian response operations. Our BERT-based NER model demonstrated superior overall accuracy with an F1-score of 0.93, outperforming the GazPNE model's F1-score of 0.90 in location entity recognition. While the BERT model achieved higher precision (0.93) and balanced performance, the GazPNE model excelled in recall (0.97) and processing speed, completing the same evaluation in 7.03 seconds compared to BERT's 114.82 seconds.

Geocoding evaluation on real Telegram message data revealed practical challenges for both approaches. The BERT model extracted 92 location entities and successfully geocoded 26 to correct coordinates, while GazPNE's high recall resulted in excessive false positives (2,391 extractions), though it still resolved 20 locations correctly. Error analysis identified several areas for improvement: expanding the gazetteer to include street-level features, incorporating more training examples of fine-grained locations such as specific addresses, and refining disambiguation strategies to handle common nouns that coincidentally match place names.

Future work will focus on expanding training datasets to include diverse location types, particularly addresses and multi-word place names. Additionally, developing hybrid approaches that combine BERT's precision with GazPNE's recall characteristics could yield more robust performance for humanitarian applications. The methods developed in this study provide a foundation for spatial analysis of displaced population communication data, enabling humanitarian organizations to identify geographic patterns in community needs and allocate resources more effectively.

References

[1] Aldana-Bobadilla, E., Molina-Villegas, A., Lopez-Arevalo, I., Reyes-Palacios, S., Muñiz-Sanchez, V., & Arreola-Trapala, J. (2020). Adaptive Geoparsing Method for Toponym Recognition and Resolution in Unstructured Text. *Remote Sensing*, 12(18), 3041. https://doi.org/10.3390/rs12183041

[2] Bashir, S. R., Raza, S., & Misic, V. B. (2023). BERT4Loc: BERT for Location—POI Recommender System. *Future Internet*, 15(6), 213. https://doi.org/10.3390/fi15060213

[3] GeoNames. (n.d.). GeoNames Geographical Database. https://www.geonames.org/

[4] Hu, X., Zhou, Z., Li, H., Hu, Y., Gu, F., Kersten, J., Fan, H., & Shimizu, K. (2023). Location Reference Recognition from Texts: A Survey and Comparison. *ACM Computing Surveys*, 56(5), 1–37. https://dl.acm.org/doi/full/10.1145/3625819

[5] John, N. (2022). CoNLL Locations Only Dataset. Kaggle. https://www.kaggle.com/datasets/nikhiljohnk/conll-locations-only

[6] MapTiler AG. (2025). Place names from OpenStreetMap. OSMNames.org. https://osmnames.org/about/

[7] Uhuohuy. (n.d.). GazPNE: Annotation-free Deep Learning for Place Name Extraction from Microblogs Leveraging Gazetteer and Synthetic Data by Rules. GitHub repository. https://github.com/uhuohuy/GazPNE